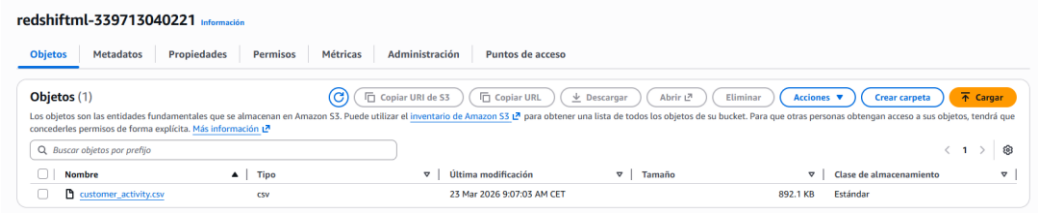
Redshift ML Training

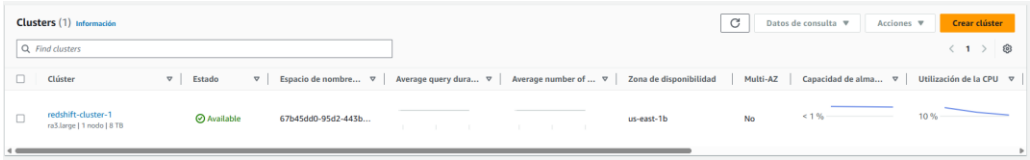
A. Requisitos previos para habilitar Amazon Redshift ML

1) General

a) Creamos el bucket redshiftml-339713040221



b) Creamos el cluster



c) Creamos usuario y sus permisos

```
create user demouser with password 'Password1';

GRANT CREATE MODEL TO demouser;
```

2) Carga de datos

a) Creamos el esquema y la tabla

```
CREATE SCHEMA demo_ml;

CREATE TABLE demo_ml.customer_activity (
    state varchar(2),
    account_length int,
    area_code int,
    phone varchar(8),
    intl_plan varchar(3),
    vMail_plan varchar(3),
    vMail_message int,
    day_mins float,
    day_calls int,
    day_charge float,
    total_charge float,
    eve_mins float,
    eve_calls int,
    eve_charge float,
    night_mins float,
    night_calls int,
    night_charge float,
    intl_mins float,
    intl_calls int,
    intl_charge float,
    cust_serv_calls int,
    churn varchar(6),
    record_date date
);
```

b) Cargamos la tabla

```
COPY demo_ml.customer_activity
FROM 's3://redshift-downloads/redshift-ml/customer_activity/'
IAM_ROLE 'arn:aws:iam::339713040221:role/LabRole'
delimiter ','
IGNOREHEADER 1
region 'us-east-1';
```

c) Añadimos permisos al usuario

```
GRANT SELECT on demo_ml.customer_activity TO demouser;

GRANT CREATE, USAGE ON SCHEMA demo_ml TO demouser;
```

B. Cree y entrene su primer modelo de ML

1) Creamos el modelo

Name	Value
Updated Rows	0
Execute time	2,619s
Start time	Mon Mar 23 09:57:37 CET 2026
Finish time	Mon Mar 23 09:57:40 CET 2026

Query	CREATE MODEL demo_ml.customer_churn_model
	FROM (
	  SELECT
	  churn,
	  state,
	  area_code,
	  account_length,
	  CASE WHEN intl_plan = 'yes' THEN 1 ELSE 0 END as has_intl_plan,
	  CASE WHEN vmail_plan = 'yes' THEN 1 ELSE 0 END as has_vmail_plan,
	  vmail_message,
	  cust_serv_calls,
	  (cust_serv_calls * 30.0 / (account_length + 0.01)) AS monthly_serv_calls_rate,
	  total_charge,
	  (total_charge / (account_length + 0.01)) as avg_daily_spend,
	  (intl_charge / (total_charge + 0.01)) as intl_spend_ratio,
	  (day_mins + eve_mins + night_mins) as total_mins,
	  (day_mins / (day_mins + eve_mins + night_mins + 0.01)) as day_usage_ratio
	  FROM demo_ml.customer_activity
	  WHERE record_date < '2020-01-01'
	)
	TARGET churn
	FUNCTION predict_customer_churn
	IAM_ROLE 'arn:aws:iam::339713040221:role/LabRole'
	SETTINGS (
	  S3_BUCKET 'redshiftml-339713040221',
	  MAX_RUNTIME 3600
	)

2) Comprobamos su creación

```
SHOW MODEL ALL;
SHOW MODEL demo_ml.customer_churn_model;
```

esultados 1 | Resultados 1 (2)

HOW MODEL ALL; SHOW MODEL demo_ml.customer_churn_model | Data filter is not supported

	Key	Value
1	Model Name	customer_churn_model
2	Schema Name	demo_ml
3	Owner	awsuser
4	Creation Time	Mon, 23.03.2026 08:57:37
5	Model State	TRAINING

3) Tras un rato terminó

```
SHOW MODEL demo_ml.customer_churn_model;
```

esultados 1

HOW MODEL demo_ml.customer_churn_model | Enter a SQL expression to filter results (use Ctrl+Space)

	Key	Value
1	Model Name	customer_churn_model
2	Schema Name	demo_ml
3	Owner	awsuser
4	Creation Time	Mon, 23.03.2026 08:57:37
5	Model State	READY
6	Training Job Status	MaxAutoMLJobRuntimeReached
7	validation:f1_binary	0.819780
8	Estimated Cost	14.917033
9		
10	TRAINING DATA:	
11	Query	SELECT CHURN, STATE, AREA_CODE, ACCOUNT_LENGTH, CASE WHEN INTL_PLAN = 'YES' THEN 1 ELSE 0 END AS HAS_INTL_PLAN, CAS
12		FROM DEMO_ML.CUSTOMER_ACTIVITY
13		WHERE RECORD_DATE < '2020-01-01'
14	Target Column	CHURN
15		
16	PARAMETERS:	
17	Model Type	xgboost
18	Problem Type	BinaryClassification
19	Objective	F1
20	AutoML Job Name	redshiftml-20260323085737505777
21	Function Name	predict_customer_churn
22		predict_customer_churn_prob
23	Function Parameters	state area_code account_length has_intl_plan has_vmail_plan vmail_message cust_serv_calls monthly_serv_calls_rate total_charge avg_dai
24	Function Parameter Types	varchar int4 int4 int4 int4 int4 int4 numeric float8 float8 float8 float8 float8
25	IAM Role	arn:aws:iam::339713040221:role/LabRole
26	S3 Bucket	redshiftml-339713040221
27	Max Runtime	3600

C. Cree y entrene su primer modelo de ML

1) Probamos

	accountid	churn	predicted
1	408411-7778	True.	False.
2	510397-6789	True.	False.
3	415395-6195	True.	False.
4	510408-4529	True.	False.
5	510329-2786	True.	False.
6	415389-2540	False.	True.
7	510399-3083	True.	False.
8	408405-4420	False.	True.
9	510397-1766	True.	False.
10	510399-7268	True.	False.
11	510365-7774	True.	False.
12	415415-4564	True.	False.
13	415333-9002	True.	False.
14	415369-1929	True.	False.
15	510373-3959	True.	False.
16	415411-8956	False.	True.

2) Modificación la consulta anterior para obtener las probabilidades de predicción.

	accountid	churn	predicted
1	415371-7191	False.	{"probabilities":[0.99964970,0.00035032],"labels":["False.","True."]}
2	408375-9999	False.	{"probabilities":[0.99908239,0.00091764],"labels":["False.","True."]}
3	510391-8027	False.	{"probabilities":[0.99460918,0.00539085],"labels":["False.","True."]}
4	510355-9993	False.	{"probabilities":[0.99239194,0.00760807],"labels":["False.","True."]}
5	415329-6603	True.	{"probabilities":[0.99917978,0.00082022],"labels":["True.","False."]}
6	415344-9403	False.	{"probabilities":[0.99955207,0.00044794],"labels":["False.","True."]}
7	408363-1107	False.	{"probabilities":[0.99979603,0.00020396],"labels":["False.","True."]}
8	510394-8006	False.	{"probabilities":[0.99995750,0.00004252],"labels":["False.","True."]}
9	415366-9238	False.	{"probabilities":[0.99552137,0.00447862],"labels":["False.","True."]}
10	510386-2923	False.	{"probabilities":[0.99887180,0.00112822],"labels":["False.","True."]}
11	510356-2992	False.	{"probabilities":[0.99889553,0.00110449],"labels":["False.","True."]}
12	415373-2782	False.	{"probabilities":[0.99927247,0.00072755],"labels":["False.","True."]}
13	415396-5800	False.	{"probabilities":[0.99986869,0.00013128],"labels":["False.","True."]}
14	415358-1958	False.	{"probabilities":[0.99653471,0.00346530],"labels":["False.","True."]}
15	415350-2565	False.	{"probabilities":[0.99986494,0.00013506],"labels":["False.","True."]}
16	415331-3698	False.	{"probabilities":[0.99975693,0.00024307],"labels":["False.","True."]}
17	408357-3817	False.	{"probabilities":[0.97751659,0.02248342],"labels":["False.","True."]}
18	415353-2630	False.	{"probabilities":[0.99921054,0.00078949],"labels":["False.","True."]}
19	510410-7789	False.	{"probabilities":[0.99977416,0.00022582],"labels":["False.","True."]}
20	415416-8428	False.	{"probabilities":[0.99970287,0.00029713],"labels":["False.","True."]}
21	510370-3359	False.	{"probabilities":[0.99930650,0.00069352],"labels":["False.","True."]}
22	415362-1407	False.	{"probabilities":[0.99951816,0.00048183],"labels":["False.","True."]}
23	415390-7274	False.	{"probabilities":[0.99950463,0.00049540],"labels":["False.","True."]}
24	408364-1995	False.	{"probabilities":[0.99899739,0.00100263],"labels":["False.","True."]}
25	408413-4957	False.	{"probabilities":[0.99987173,0.00012829],"labels":["False.","True."]}
26	408420-5645	False.	{"probabilities":[0.99973816,0.00026182],"labels":["False.","True."]}
27	415404-3211	False.	{"probabilities":[0.99952561,0.00047437],"labels":["False.","True."]}
28	415353-3759	True.	{"probabilities":[0.99207062,0.00792938],"labels":["True.","False."]}
29	415340-5121	False.	{"probabilities":[0.99993730,0.00006270],"labels":["False.","True."]}
30	415403-9733	False.	{"probabilities":[0.99945921,0.00054079],"labels":["False.","True."]}

D. Invoca tu modelo ML para la inferencia

1) Predicción de qué clientes del código de área 408 podrían darse de baja.

accountid	churn	predictedactive
408335-4719	False.	False.
408350-8884	False.	False.
408393-7984	True.	True.
408418-6412	False.	False.
408383-1121	False.	False.
408360-1596	True.	True.
408395-2854	False.	False.
408341-9764	False.	False.
408332-9891	False.	False.
408372-9976	False.	False.
408383-6029	True.	True.
408353-3061	False.	False.
408349-4396	False.	False.
408363-5947	False.	False.
408370-7574	True.	False.
408355-7251	False.	False.
408359-5893	False.	False.
408365-9011	False.	True.
408338-9472	False.	False.
408359-1231	False.	False.
408360-1171	True.	True.
408366-4467	False.	False.
408401-7594	False.	False.
408397-4976	False.	False.
408334-2577	False.	False.
408403-4298	False.	False.
408409-3786	False.	False.
408383-1509	False.	False.

2) Probabilidades de predicción de la salida de etiquetas para el escenario anterior.

	probabilities	count
1	0,9	3.214
2	0,8	47
3	0,7	30
4	0,6	25
5	0,5	17

3) Calcular el recuento total de probabilidades de predicción en todos los datos de inferencia para determinar el rango de valores que podría ayudar a establecer el umbral de uso de la etiqueta predicha.

	123 probabilities	123 count
1	0,9	3.214
2	0,8	47
3	0,7	30
4	0,6	25
5	0,5	17